BIBO: is an AI-powered and responsive E-learning System designed for Out-of-School Youth ages 7 to 12.

A Thesis Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Bacoor

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Science

Kelvin C. Parillo
Blessie Mae B. Bardaje
John Paulo P. Silo
Rodrigo A. Amparado Jr.

November 29, 2025

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **BIBO: is an AI-powered and responsive E-learning System designed for Out-of-School Youth ages 7 to 12.**

NAME OF PROPONENTS: Kelvin C. Parillo
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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Computer Science
has been examined and is recommended for Proposal Defense.

ENDORSED BY:

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November 29, 2025

APPROVAL SHEET

This thesis proposal titled **BIBO: is an AI-powered and responsive e-learning system designed for out-of-school youth ages 7 to 12.**, prepared and submitted by **Kelvin C. Parill; Blessie Mae B. Bardaje; John Paulo P. Silo**; and **Rodrigo A. Amparado Jr.**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science, has been examined and is recommended for acceptance and approval.

Mr. Nestor Miguel T. Pimentel
Thesis Adviser

Accepted and approved by the Thesis Review Panel
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science

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INTRODUCTION

Education plays a crucial role in shaping children's foundational skills; yet, many out-of-school youth in the Philippines struggle to access structured learning opportunities.

Factors such as poverty, distance from schools, learning gaps, and lack of academic support prevent young learners from continuing formal education. With the advancement of modern technology and the increasing availability of digital tools, e-learning systems have become a practical alternative for delivering accessible and flexible education. BIBO is an AI-powered and responsive e-learning system designed specifically for out-of-school youth ages 7 to 12. It integrates artificial intelligence, gamified learning, and adaptive modules aligned with foundational academic subjects. Through BIBO, learners can access interactive lessons in English, Filipino, Math, and Science on both mobile and web platforms. The system also supports teachers and parents through monitoring dashboards and structured tutorials that follow DepEd's ARAL Program.

The system aims to provide an equitable and modernized learning environment where young learners can receive personalized instruction, improve their academic competencies, and regain their motivation to continue schooling.

In recent years, digital learning has become increasingly essential in addressing educational inequalities, particularly in underserved communities where traditional learning resources are limited. Many children lack access to books, trained educators, and safe learning spaces, resulting in low literacy and numeracy rates. By utilizing artificial intelligence and gamified interfaces, digital platforms like BIBO can deliver engaging and adaptive learning experiences that traditional classrooms may not always provide. These technologies enable individualized instruction, making it possible to cater to each learner's pace, skill level, and specific learning needs.

The rise of mobile technology has opened new opportunities for implementing learning solutions beyond the classroom. Since most Filipino households have access to smartphones or shared devices, digital education becomes more feasible even in low-income settings. BIBO leverages this accessibility by ensuring its platform is lightweight, responsive, and usable on common devices. This design allows children—regardless of their socio-economic background—to interact with high-quality educational materials anytime and anywhere in Bacoor.

Additionally, parents and educators play a critical role in nurturing children's study habits and

monitoring progress. However, many lack the tools or time to provide consistent guidance. BIBO bridges this gap through its built-in analytics and monitoring dashboard, which helps adults track learners' performance and identify areas requiring support. By integrating technology with structured guidance, the system fosters a collaborative learning environment where families, teachers, and learners work together toward academic improvement.

Background of the problem

Out-of-school youth (OSY) ages 7 to 12 represent a growing concern in Bacoor City, Cavite, Philippines. Many children lack access to quality education due to financial limitations, family responsibilities, distance from schools, and learning difficulties. These issues lead to widening academic gaps, low literacy levels, and reduced opportunities for reintegration into formal schooling.

While e-learning platforms exist, most of them are not designed specifically for OSY learners who require simpler interfaces, foundational-level content, and guided, supportive instruction. Existing systems often lack AI tutoring and real-time progress monitoring, language options, and parent involvement features—which are crucial for young independent learners.

BIBO aims to address these gaps by offering an easy-to-use AI-enhanced e-learning system that delivers foundational lessons, adaptive feedback, and engaging activities to promote consistent learning. The system also empowers parents and teachers to track learner progress and provide targeted interventions. As a result, BIBO supports educational recovery and reduces learning gaps among marginalized Filipino youths in Bacoor.

Overview of the current state of the technology

Out-of-school youth (OSY) ages 7 to 12 represent a growing concern in Bacoor City, Cavite. Many children lack access to quality education due to financial limitations, family responsibilities, distance from schools, and learning difficulties. These issues lead to widening academic gaps, low literacy levels, and reduced opportunities for reintegration into formal schooling.

While e-learning platforms exist, most of them are not designed specifically for OSY learners who require simpler interfaces, foundational-level content, and guided, supportive instruction. Existing systems often lack AI tutoring and real-time progress monitoring, multi-language options, and parent

involvement features—which are crucial for young independent learners.

BIBO aims to address these gaps by offering an easy-to-use AI-enhanced e-learning system that delivers foundational lessons, adaptive feedback, and engaging activities to promote consistent learning. The system also empowers parents and teachers to track learner progress and provide targeted interventions. As a result, BIBO supports educational recovery and reduces learning gaps among marginalized Filipino youths in Bacoor City.

Objectives of the study

The primary objective of this study is to develop and evaluate **BIBO**, an AI-powered and responsive e-learning system designed to support the foundational learning needs of out-of-school youth (OSY) ages 7 to 12 in the Philippines. Specifically, this study aims to:

1. Develop an accessible and responsive e-learning platform

To design and implement a mobile and web-based learning system that provides easy-to-use interfaces suitable for young learners, regardless of their socio-economic background or device limitations.

2. Integrate AI-driven adaptive learning features

To incorporate artificial intelligence that delivers personalized lessons, adaptive feedback, and learning recommendations based on each learner's performance, pace, and identified learning gaps.

3. Provide interactive and gamified learning modules

To create engaging and curriculum-aligned activities in English, Filipino, Mathematics, and Science using gamification techniques to enhance motivation and sustain learner engagement.

4. Support parents and teachers through monitoring and analytics tools

To develop a real-time dashboard that enables parents, guardians, and teachers to monitor learner progress, identify areas of difficulty, and provide timely interventions aligned with DepEd's ARAL Program.

5. Address the unique educational needs of out-of-school youth

To ensure that the system responds specifically to the challenges faced by Out-of-School Youth—such as learning gaps, a lack of tutorials, and limited academic support—through structured foundational content and guided learning pathways.

6. Evaluate the system's usability, effectiveness, and impact

To assess how BIBO improves learners' academic competencies, encourages learning continuity, and supports educational recovery among marginalized Filipino youth through user testing and feedback.

Scope and limitations of the study

The system covers lessons in English, Math, and Filipino for learners aged 7–12 years old. Accessible via web browsers, Android devices. Includes AI tutoring, gamified quizzes, progress tracking, and teacher/parent access. Follows and inspires DepEd's ARAL Program structure for small-group tutorials. Focuses on foundational education for out-of-school youth. The AI tutor is limited to responding to basic lesson-related queries. Not intended to replace formal schooling; serves as supplementary learning. The AI tutor is limited to responding to basic lesson-related queries, which may restrict its ability to address more complex questions that require deeper contextual understanding. Offline functionalities may be limited depending on device type.

Review of related literature/studies/systems

The review of literature and related systems highlights the importance of gamification, artificial intelligence, and adaptive learning in improving student engagement and performance. Foreign studies show that game elements and AI-powered tutoring significantly enhance motivation and personalize the learning experience. Local Philippine literature supports these findings and emphasizes the need for simple, accessible, mobile-friendly educational tools due to issues like digital inequality and limited teacher readiness.

The analysis of existing systems shows that while many platforms use games, analytics, and AI, none are specifically designed for out-of-school Filipino youth. These systems also reveal gaps in accessibility, localization, and parent–teacher monitoring features.

Overall, the chapter concludes that combining AI tutoring, gamified lessons, localized content, and progress tracking is crucial for addressing the learning gaps of OSYs aged 7–12. The proposed system, BIBO, directly addresses these needs by providing an engaging, adaptive, and culturally relevant learning

platform.

Foreign Literature

Recent international studies highlight the growing role of gamification, artificial intelligence, and adaptive technologies in improving digital learning environments. Zainuddin et al. (2020) emphasized that gamification elements—such as points, badges, progress bars, and leaderboards—greatly improve learner motivation and academic performance when integrated with clear instructional objectives. Similar findings were reported by Toda et al. (2020), who mapped global gamification trends and observed consistent gains in engagement across K–12 and higher education. However, they cautioned that overly complex game mechanics may overwhelm younger learners and stressed the importance of age-appropriate, user-centered design.

Alongside gamification, the integration of artificial intelligence has significantly enhanced digital learning systems. Lin, Huang, and Lu (2023) reviewed AI-driven Intelligent Tutoring Systems (ITS) and found that AI improves mastery and comprehension by analyzing learner behavior and providing adaptive feedback. Their study also noted that ITS strengthens student autonomy and supports sustainable, long-term education. Complementing this, Salas-Pilco et al. (2022) demonstrated that AI and learning analytics help educators identify learning difficulties early, customize interventions, and enhance decision-making through real-time progress monitoring—although ethical concerns related to data privacy remain key challenges.

A broader review of Intelligent Tutoring Systems by Alrakhawi et al. (2023) identified essential ITS components such as learner modeling, knowledge representation, and automated feedback algorithms. They concluded that ITS enhances learning accuracy and accelerates skill mastery but noted that many existing systems still focus on single-subject applications. Son (2024) further examined ITS specifically in mathematics education and found that these systems improve conceptual understanding, reduce math anxiety, and support self-paced learning. However, teacher training and technical constraints continue to limit widespread adoption.

Personalized learning has also become central to modern educational technology. Ahmed, Tlili, and Lampropoulos (2024) reported that adaptive learning systems adjust content difficulty and sequencing to meet individual learner needs, effectively reducing learning gaps—especially among struggling learners. Nonetheless, the authors recognized that many adaptive platforms require advanced digital literacy, posing accessibility challenges for younger or marginalized learners. Supporting these findings,

Strielkowski et al. (2025) argued that AI-driven adaptive learning promotes long-term educational equity by automating content adaptation, predicting learner needs, and easing teacher workload. Their study asserts that adaptive AI technologies are essential to future-ready global education.

Emerging technologies such as virtual reality have also contributed to improved learning experiences. Lampropoulos and Kinshuk (2024) found that VR, when combined with gamification, enhances immersion, engagement, and long-term retention by providing experiential learning environments. Despite these advantages, high costs, technical complexity, and limited accessibility remain barriers to large-scale implementation. The authors highlight the need for future VR-enabled learning systems to adopt more inclusive, accessible, and user-centered design approaches.

Overall, foreign literature collectively demonstrates that gamification, AI-powered tutoring, adaptive learning, and immersive media significantly enhance learner engagement, personalization, and academic performance. These findings provide strong support for integrating similar technologies into educational platforms designed for young learners, particularly those with limited access to formal schooling.

Local Literature

Local studies in the Philippines highlight a growing interest in gamification, adaptive learning, and the integration of digital technologies to improve student engagement and learning outcomes. In mathematics education, Nob, Roble, and Lomibao (2024) found that gamified strategies improve motivation and performance among Filipino learners by incorporating rewards, competition, and progress tracking. However, they noted that many teachers lack sufficient training to implement gamification effectively. Similarly, Luzano (2024) reported that gamified instruction in tertiary mathematics enhances engagement, reduces math anxiety, and improves conceptual retention, though inconsistent implementation and limited institutional support remain challenges.

Gamification has also shown positive effects in literacy and other subject areas. Doria et al. (2025) demonstrated that gamified reading activities significantly improve comprehension among Grade 4 pupils, with narrative-based quests increasing reading interest. However, they observed that excessive competition can sometimes hinder collaboration. In Technology and Livelihood Education, Pagente et al. (2024) found that game-based tasks increase motivation and completion rates, although technological limitations in public schools hinder widespread adoption. The researchers recommend designing low-bandwidth or offline gamified tools to address these barriers.

Beyond K–12 classrooms, gamification has been applied to diverse learning contexts. Argenal et al.

(2024) reported enhanced engagement, comprehension, and critical-thinking skills among foreign medical students exposed to gamified science and mathematics lessons. They cautioned, however, that some tasks may be too simplified for advanced learners, suggesting the need for adjustable difficulty levels. Likewise, Guisadio et al. (2025) conducted a systematic review of gamification in Philippine basic education and identified reward systems, progress indicators, and narrative missions as the most widely used elements. Although these features boost engagement, the lack of standardized frameworks results in inconsistent outcomes across schools.

Studies on flexible and hybrid learning formats echo similar benefits. Moreno, Gimpaya, and Aliazas (2025) found that gamified Hyflex physical education increases motivation and physical activity among Filipino youth, though unstable internet connectivity poses challenges. In flexible learning environments, Besavilla et al. (2024) noted that gamified applications sustain motivation and attention in remote learning, despite issues like device incompatibility and screen fatigue. The authors emphasize the need for mobile-friendly and adaptive gamified systems to support diverse learners.

Research also highlights persistent issues of digital inequity, particularly among out-of-school Filipino youth. Manila, Reyes, and Dizon (2023) observed that limited access to devices, financial constraints, and poor connectivity continue to restrict participation in digital learning. Many marginalized learners rely on borrowed or shared devices, limiting their study time and engagement. These findings underscore the necessity of inclusive, accessible, and low-resource digital learning platforms for underserved communities.

Meanwhile, studies on artificial intelligence in Philippine education show both promise and barriers. Dizon and Castro (2025) found that while teachers express interest in using AI tools, they face challenges related to inadequate infrastructure, lack of training, and concerns about job displacement. Despite these barriers, properly utilized AI tools have been shown to improve instructional efficiency. The authors recommend targeted capacity-building initiatives to support AI integration across K–12 instruction.

Overall, local literature reveals strong evidence that gamification and emerging technologies can enhance learning in the Philippines, but also emphasizes the need for accessible systems, teacher training, standardized frameworks, and inclusive digital tools—especially for marginalized and out-of-school youth.

Related System

Modern digital learning platforms now integrate features such as gamification, artificial intelligence,

adaptive learning, and data analytics to boost learner engagement and academic performance. Existing studies describe how these systems operate, the challenges they address, and the opportunities they offer for improving emerging platforms like BIBO.

Gamified learning environments have been shown to significantly enhance learner motivation and achievement. According to Dikmen et al. (2020), systems that incorporate rewards, challenges, and performance monitoring help improve academic outcomes. Their findings highlight that user-friendly interfaces strengthen engagement, but personalization often remains limited due to weak adaptive mechanisms. To overcome this, they recommend combining gamification with adaptive learning algorithms to support varying learner profiles. In a similar vein, Facer and Williamson (2011) studied mobile gamified applications such as Duolingo and found them effective in supporting literacy development through reward-based progress systems. However, they noted shortcomings in multilingual support and cultural localization, suggesting the need for AI-enhanced customization especially for non-native speakers.

Systems tailored for specialized learners further emphasize the advantages of game-based instruction. Tlili et al. (2022) observed that platforms designed for learners with disabilities improve accessibility, motivation, and personalized assistance. Although activity theory can guide inclusive design, many systems remain inadequately optimized for different types of impairments. They call for more adaptive and inclusive digital environments to cater to diverse users.

Adaptive learning systems used in K–12 settings also demonstrate the impact of data-driven customization. Chen and Huang (2022) examined platforms like DreamBox and ALEKS, finding that these tools adjust content based on individual learner data, leading to better mastery in math and science. Despite their effectiveness, these platforms encounter challenges related to data privacy and accessibility in low-resource communities. The authors recommend hybrid offline–online models to expand access. Meanwhile, Radianti et al. (2020) reported that immersive virtual reality tools such as Google Expeditions enhance experiential learning, but high hardware costs and technical limitations hinder widespread adoption—particularly for out-of-school youth. Low-cost, lightweight VR solutions are proposed to address this gap.

Artificial intelligence has also pushed advancements in digital tutoring and personalized support. Winkler and Soellner (2018) found that AI chatbots powered by systems like IBM Watson can provide immediate assistance and tailored responses. Nonetheless, their difficulty in understanding complex or context-specific questions remains a major limitation, underscoring the need for more multilingual and

context-aware systems. Hariyanto et al. (2025) further showed that AI-powered adaptive platforms adjust learning pathways based on user behavior, improving mastery and engagement. However, concerns regarding algorithm transparency and data privacy highlight the importance of ethical and explainable AI.

Instructional design frameworks also play a vital role in strengthening gamified learning platforms. González-Fernández et al. (2022) stressed that systematic instructional design leads to better usability, consistent learning experiences, and improved alignment between gamified elements and academic objectives. They caution that poorly executed gamification can negatively affect learning, emphasizing the importance of evidence-based design.

In areas with limited internet access, offline and hybrid learning systems remain crucial. Trucano (2005) noted that offline e-learning solutions using local servers and portable devices are valuable in rural settings. While these systems support gamified content, they struggle with updating materials and scaling distribution. Hybrid systems that allow offline access with periodic online synchronization are recommended, especially for underserved learners like out-of-school youth.

Learning analytics dashboards are also essential in monitoring learner progress. Papamitsiou and Economides (2014) analyzed dashboards in platforms such as Blackboard and found that they effectively identify learners who may need intervention. Although privacy concerns persist, well-designed dashboards—with features like parent-access options—help strengthen instruction and home-based learning.

METHODOLOGY

The developmental research approach focuses on the design, creation, and evaluation of the BIBO system. The research includes the analysis, design, development, testing, and evaluation stages. This study also adopts the Agile Scrum framework, supported by the system's Entity– Relationship Diagram (ERD) and initial Drawing Board/Wireframe drafts as part of its iterative design process.

The Agile Scrum methodology is used to ensure that the system is developed in flexible, incremental cycles called sprints. Each sprint involves continuous collaboration among developers, researchers, and evaluators to refine system features based on feedback. Through this iterative approach, improvements can be made quickly, ensuring that the BIBO platform remains responsive to the learning needs of out-of-school youth. Agile Scrum also supports efficient task management by breaking down complex system functions into manageable modules, allowing the team to focus on delivering usable components at the end of each cycle.

As part of the design phase, the Entity–Relationship Diagram (ERD) is created to illustrate the structure of the system's database, including its entities, attributes, and relationships. This ensures that all data—such as learner profiles, lesson modules, performance analytics, and quiz results—is organized logically and securely.

Hardware/Software

- Desktop or laptop for development
- Mobile devices (Android/iOS) for testing
- Server hosting
- Stable internet connection
- Programming languages: HTML, CSS, JavaScript, PHP, Python

- Frameworks: React Native / Flutter for mobile, Laravel / Node.js for web
- Database: MySQL / Firebase
- AI tools: NLP model for chatbot
- Development tools: VS Code, GitHub, Figma

Calendar of Activities

Activity	Duration	Description
Requirements Gathering	Week 1–2	Identify needs, features
System Design	Week 3–5	UI/UX, database design

Development Week 6-12 **Web+Mobile system Creation**

AI Tutor Integration Week 10-14 **NLP Model Trainig**

Testing C Quality Assurance
Week 15-17 **Bug Fixing, Revisions**

Evaluation Week 19 **User Testing , Feedback**

Budgetary Estimate

The budgetary estimate outlines the projected costs required to design, develop, test, and deploy the BIBO AI-powered e-learning system. The expenses include hardware, software, development tools, cloud services, content creation, and operational costs necessary for the implementation of the system.

Budgetary Estimate

Quantity	Specifics	Approximate Cost

Human Resources

The following pages contain the curriculum vitae of the researchers and the Adviser's Acceptance Form.

Curriculum Vitae of

Kelvin C. Parillo

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Tikyo993@gmail.com
09280745290

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2021-present	STI College Bacoor
Vocational/Technical	2020-2021	Progresive Senior High School
High School	2017-2028	General Emilio Aguinal National High School
Elementary	2012-2013	Arakan Central Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of

<GIVEN NAME MI. FAMILY NAME>

<complete address>
<email address>
contact number, either cellular phone or landline or both

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary		
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Inclusive Dates	Title of Training, Seminar or Workshop
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Listed in reverse chronological order (most recent

ADVISER'S ACCEPTANCE FORM

NAME OF PROPONENTS: **Kelvin C. Parillo**
Blessie Mae B. Bardaje
Rodrigo A. Amparado Jr.
John Paulo P. Silo

APPROVED RESEARCH TITLE: **BIBO: is an AI-powered and responsive e-learning system designed for out-of-school youth ages 7 to 12.**

AREA OF STUDY: **<Area of Research and Field of Study>**

CONFORME:

Mr. Nestor Miguel Pimentel
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APPROVED BY:

Prof. Rocelle U. Camposagrado, MIT **Date:**
Thesis Coordinator

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